Calculus I 2016 Final Solutions

1. Find/evaluate the following integrals.

   (a) $\int_0^\pi \sin \frac{x}{3}\,\mathrm{d}x$

   (b) $\int x \tan^{-1} x\,\mathrm{d}x$

   (c) $\int_{-\pi}^{\pi} \theta \sin^2 \theta e^{\theta^2}\,\mathrm{d}\theta$

Solution

(a) $= [-3\cos\frac{x}{3}]_0^\pi = -3(1/2-1) = 3/2$

(b) Letting $f(x) = \tan^{-1} x$ and $g'(x) = x$ we have that $f'(x) = \frac{1}{1+x^2}$ so using Int by parts

$$\begin{aligned}\int x\tan^{-1} x\,\mathrm{d}x &= \frac{x^2}{2}\tan^{-1} x - \int \frac{x^2}{2(1+x^2)}\,\mathrm{d}x \\ &= \frac{x^2}{2}\tan^{-1} x - 1/2\int 1\,\mathrm{d}x + 1/2\int \frac{1}{1+x^2}\,\mathrm{d}x \\ &= \frac{x^2}{2}\tan^{-1} x - \frac{x}{2} + \tan^{-1} x + C\end{aligned}$$

(c) 0: the function is odd and the interval of integration is symmetric about 0.

2. Find the indefinite integral $\int \tan^2 x \sec^2 x\,\mathrm{d}x$.

Solution

The power 2 of sec is even, so we use the substitution $u = \tan x$. This gives $\mathrm{d}u = \sec^2 x\,\mathrm{d}x$ so

$$\int \tan^2 x \sec^2 x\,\mathrm{d}x = \int u^2\,\mathrm{d}u = \frac{u^3}{3} + C = \frac{\tan^3}{3} + C$$

3. Determine if the following improper integrals converge, and if they do, evaluate them.

   (a) $\int_0^1 x^{-1/2}\,\mathrm{d}x$.

   (b) $\int_0^{\pi/2} \tan x\,\mathrm{d}x$.

Solution

(a) Converges: $\int_0^1 x^{-1/2}\,\mathrm{d}x = \lim_{t\to 0}\int_t^1 x^{-1/2}\,\mathrm{d}x = \lim_{t\to 0}[2x^{1/2}]_t^1 = 2\lim_{t\to 0}(1-\sqrt{t}) = 2$

(b) Recalling that $\int \tan x\,\mathrm{d}x = -\ln|cosx| + C$, (or using the substitution $u = \cos x$ to show it:

$$\int \frac{\sin x}{\cos x}\,\mathrm{d}x = -\int \frac{1}{u}\,\mathrm{d}u = -\ln|u| = -\ln|\cos x| + C)$$

we get that

$$\int_0^{\pi/2} \tan x\,\mathrm{d}x = \lim_{t\to\pi/2}\int_0^t \tan x\,\mathrm{d}x = \lim_{t\to\pi/2} -\ln\cos t = \infty$$

so this diverges to infinity.

4. Find the volume of the solid created by rotating the region bounded by $y^2 = x$ and $x = 3y$ about the y-axis.

Solution

Setting $y^2 = x = 3y$ and solving we see the curves intersect at at $y = 0, 3$, so $x = 0, 9$. The intersections are both for non-negative y so the region we are rotating is bounded by $y = x/3$ and the positive root $y = \sqrt{x}$ of the upper curve. By the method of shells we get that the volume is

$$2\pi\int_0^9 x(\sqrt{x} - x/3)\,\mathrm{d}x = 2\pi\left[\frac{2}{5}x^{\frac{5}{2}} - \frac{x^3}{9}\right]_0^9 = 2\pi\left(\frac{2\cdot 243}{5} - 81\right) = \frac{162\pi}{5}.$$

By the method of slices we get

$$\pi\int_0^3 (9y^2 - y^4)\,\mathrm{d}y = \pi\left[3y^3 - \frac{y^5}{5}\right]_0^3 = \pi(81 - \frac{243}{5}) = \frac{162\pi}{5}.$$

5. Find the length of the curve $y = \ln(1 - x^2)$ on the interval $0 \le x \le 1/2$.

Solution

Finding $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{-2x}{1-x^2}$ we get that the arc differential is

$$\mathrm{d}s = \sqrt{1 + \frac{4x^2}{(1-x^2)^2}} = \sqrt{\frac{1 - 2x^2 + x^4 + 4x^2}{(1-x^2)^2}} = \sqrt{\frac{(1+x^2)^2}{(1-x^2)^2}} = \frac{1+x^2}{1-x^2} = -1 + \frac{2}{1-x^2}$$

So the Arc Length is

$$-\int_0^{1/2} 1\,\mathrm{d}x + \int_0^{1/2} \frac{2}{1-x^2}\,\mathrm{d}x = -\frac{1}{2} + \int_0^{1/2} \frac{2}{1-x^2}\,\mathrm{d}x$$

Most people finished this off using partial fractions:

$$\int_0^{1/2} \frac{2}{1-x^2}\,\mathrm{d}x = \int_0^{1/2} \frac{1}{1+x}\,\mathrm{d}x - \int_0^{1/2} \frac{1}{1-x}\,\mathrm{d}x = [\ln|1+x| - \ln|1-x|]_0^{1/2} = \ln 3$$

Getting a final solution of $\ln 3 - 1/2$.
Some people finished it off with a substitution. Letting $x = \sin\theta$ we get

$$\int_0^{1/2} \frac{2}{1-x^2}\,\mathrm{d}x = 2\int_0^{\pi/6} \frac{\cos\theta}{\cos^2\theta}\,\mathrm{d}\theta = 2\int_0^{\pi/6} \sec\theta\,\mathrm{d}\theta = 2[\ln|\tan\theta + \sin\theta|]_0^{\pi/6} = \ln 3$$

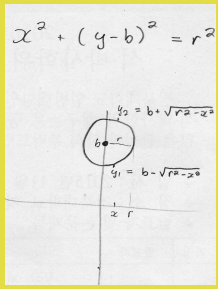
6. Find the surface area of the solids made by rotating the curve

$$(x-a)^2 + (y-b)^2 = r^2$$

(for $b > r$) around the x-axis.

Solution

We first observe that the surface area is unchanged by setting $a = 0$; the figure is then symmteric around the y-axis so we will integrate from $x = 0$ to r (instead of $-r$ to r).
For each value of x, there are two parts of the surface that we make by rotating around the x-axis. One at $y_1 = b + \sqrt{r^2 - x^2}$ and one at $y_2 = b - \sqrt{r^2 - x^2}$.



These sum to $2b$, and the slopes have the same magnitude at the two values of y so the arc differentials Δs are the same, so we can integrate $2\int_0^r 2\pi(2b)\Delta s\,\mathrm{d}x$ instead of $2\int_0^r 2\pi y_1 \Delta s\,\mathrm{d}x + 2\int_0^r 2\pi y_2 \Delta s\,\mathrm{d}x$.
Using implicit differentiation to find

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{-2x}{2(y-b)} = \frac{-x}{y-b} = \frac{-x}{\sqrt{r^2 - x^2}}$$

(here we have used the upper value y_1 of y so the b cancels), we get arc differential

$$\mathrm{d}s = \sqrt{1 + \frac{x^2}{r^2 - x^2}} = \frac{r}{\sqrt{r^2 - x^2}}.$$

Using substitituion $x = r\sin\theta$ so $\mathrm{d}x = r\cos\theta\,\mathrm{d}\theta$ we get that the surface area is

$$\begin{aligned} 2\int_0^r 2\pi(2b)\frac{r}{\sqrt{r^2 - x^2}}\,\mathrm{d}x &= 8br\pi\int_0^r \frac{1}{\sqrt{r^2 - x^2}}\,\mathrm{d}x \\ &= 8br\pi\int_0^{\pi/2} \frac{1}{r\cos\theta} r\cos\theta\,\mathrm{d}\theta \\ &= 8br\pi\int_0^{\pi/2} 1\,\mathrm{d}\theta = 4br\pi^2 \end{aligned}$$

7. Find the area enclosed by the positive x-axis and the curve $r = \theta$ ...

   (a) from $\theta = 0$ to $\theta = 2\pi$.

   (b) from $\theta = 0$ to $\theta = 4\pi$.

Solution

Here is a sketch of the regions:

The area for the first is

$$\frac{1}{2}\int_0^{2\pi} r^2\,\mathrm{d}\theta = \frac{1}{2}\int_0^{2\pi}\theta^2\,\mathrm{d}\theta = \frac{1}{2}\frac{(2\pi)^3}{3} = \frac{4}{3}\pi^3.$$

For the second,

$$\frac{1}{2}\int_{2\pi}^{4\pi}\theta^2\,\mathrm{d}\theta = \frac{1}{2}\frac{(4\pi)^3}{3} - \frac{4}{3}\pi^3 = \frac{32}{3}\pi^3 - \frac{4}{3}\pi^3 = \frac{28}{3}\pi^3.$$

(Careful: you can't integrate over $\int_0^{4\pi}$. This counts the first region twice.)

8. Find the arc length of the curve $r = 4\sin\theta$ on the interval $0 < \theta < \pi/2$.

Solution

The arc differential in polar coordinates is $\mathrm{d}s = \sqrt{r^2 + (\frac{\mathrm{d}r}{\mathrm{d}\theta})^2}\,\mathrm{d}\theta$ and $\frac{\mathrm{d}r}{\mathrm{d}\theta} = 4\cos\theta$ so the arc length is

$$\int_0^{\pi/2}\sqrt{(4\sin\theta)^2 + (4\cos\theta)^2}\,\mathrm{d}\theta = \int_0^{\pi/2} 4\,\mathrm{d}\theta = 2\pi.$$

9. Show that if $\sum a_i$ converges but $\sum b_i$ doesn't then $\sum(a_i + b_i)$ doesn't.

Solution

We prove the contrapositive: if $\sum a_i$ converges to some A, and $\sum(a_i + b_i)$ converges to some C then $\sum b_i$ converges to $C - A$.
Indeed, if $\sum a_i$ and $\sum(a_i + b_i)$ converge, then we can add or subtract them termwise, and get

$$\sum b_i = \sum(a_i + b_i) - a_i = \sum(a_i + b_i) - \sum a_i = C - A.$$

(Note, we saw in class that you cannot add series termwise if the do not converge. Many people tried to do this.)

10. Determine the absolute convergence of the following series.

(a) $\sum_{n=1}^{\infty} \frac{cos^2 n}{n^2+2}$

(b) $\sum_{n=1}^{\infty} \frac{n+2^n}{n+4^n}$

(c) $\sum (-1)^n \frac{3}{\sqrt{n}+1}$

Solution

(a) $\sum_{n=1}^{\infty} \frac{cos^2 n}{n^2+2} < \sum_{n=1}^{\infty} \frac{1}{n^2}$ which converges by the p-series test (as $p = 2 > 1$) so the original series converses by the comparison test. (As it is positive, it converges absolutley).

(b) It converges. Indeed $\sum \frac{2^n}{4^n} = \sum \left(\frac{2}{4}\right)^n$ converges by the p-series test and as

$$\lim_{n\to\infty} \frac{\frac{2^n}{4^n}}{\frac{n+2^n}{n+4^n}} = \lim \frac{2^n}{n+2^n} \frac{n+4^n}{4^n} = 1 \cdot 1 = 1$$

we have by the limit comparison text that $\sum \frac{n+2^n}{n+4^n}$ converges.

(As it is positive, it converges absolutley).

(c) This alternates and decreases so converges by the alternating series test, it doesn't converge absolutely though as $\lim_{n\to\infty} \frac{\frac{3}{\sqrt{n}+1}}{\frac{1}{\sqrt{n}}} = 3$ and $\sum \frac{1}{\sqrt{n}} = \sum \frac{1}{n^{1/2}}$ diverges by the p-series test.

11. Find the raduis of convergence and interval of convergence of $\sum_{n=0}^{\infty} \frac{x^n}{\ln(n+2)}$.

Solution

$\lim_{n\to\infty} \frac{\ln(n+3)}{\ln(n+2)} = 1$ so the radius of convergences is $R = 1$. When $x = -1$ the series is alterningt and decreasing so converges. When $x = 1$ the series is $\sum \frac{1}{\ln n+2} > \sum \frac{1}{n^{1/2}}$ which diverges, so diverges. The interval of convergence is $[-1, 1)$.

12. Find the Taylor expansion of $f(x) = x^{-2}$ about $a = 2$. What is its radius of convergence.

Solution

Computing a couple of derivives:

$$f'(x) = (-2)x^{-3}, \qquad f^{(2)}(x) = (6)x^{-4}, \qquad f^{(n)} = (-1)^n(n+1)!x^{-2-n}$$

we get $f^{(n)}(2) = (-1)^n(n+1)!2^{-2-n}$. The Taylor series is thus

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(2)}{n!}(x-2)^n = \sum (-1)^n \frac{(n+1)!}{n!2^{n+2}}(x-2)^n = \sum (-1)^n \frac{n+1}{2^{n+2}}(x-2)^n.$$

As $\lim_{n\to\infty} \frac{(n+2)2^{n+2}}{2^{n+3}n+1} \to 1/2$ the radius of convergence is 2.